

Question 10

Max. score: 4.00

Which of the following statements about GloVe is true?

- ☐ GloVe represents words as dense vectors by modeling the relationships between words using only local contexts.
- ☐ The loss function in GloVe is designed to minimize the squared difference between predicted and actual co-occurrence probabilities.
- ☐ GloVe uses a fully connected neural network to learn word vectors from a co-occurrence matrix.
- ☐ GloVe and Word2Vec use identical algorithms for generating word embeddings.

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Question 11

Max. score: 4.00

ABC is a food aggregator that is advertising that it can deliver an order in 40 minutes or less on average providing that the delivery location is within a 5 km radius of the restaurant where the order is placed.

The mean delivery time is 42.3 and the p-value is 2.5×10^{-6} .

Assume that the delivery distribution is approximately normal. Which of the following statements about this scenario is correct:

Statements:

1. The null hypothesis is rejected at a 5% level of significance. Hence, at a 5% level of significance, there is not enough evidence that the mean delivery time within a 5 km radius is actually 40 min or less, as claimed by ABC in their advertisement.
2. The null hypothesis is accepted at a 5% level of significance. Hence, at a 5% level of significance, there is enough evidence that the mean delivery time within a 5 km radius is actually 40 min or less, as claimed by ABC in their advertisement.
3. The null hypothesis is rejected at a 10% level of significance. Hence, at a 10% level of significance, there is enough evidence that the mean delivery time within a 5 km radius is actually 40 min or less, as claimed by ABC in their advertisement.

☐ 1

☐ 2

☐ 3

☐ It cannot be determined

Question 12

Max. score: 4.00

In Machine Learning, you are working on a project using a neural network. You are using the rectified linear (ReLU) activation function. Which of the following statements about the ReLU activation function is correct:

1. It is a piecewise linear function
2. If it is positive, it outputs the input directly. Otherwise, it outputs zero.
3. It is capable of providing a true zero value
4. It acts as a linear activation function.

☐ 1 and 2

☐ 2 and 3

☐ 3 and 4

☐ All of these

Reset Answer

Question 13

Max. score: 2.00

Artificial Neural Network (ANN) is also comprehended as Universal Function Approximators. Which of the following is one of the important reasons behind this?

☐ Weights

☐ Activation function

☐ Sum of the inputs used during training

☐ Loss

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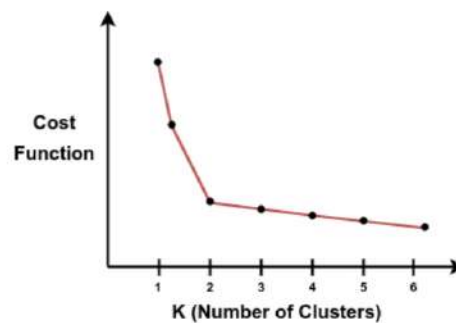
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Question 14

Max. score: 6.00

In Machine Learning, you are working on an unsupervised algorithm called the **k-means** algorithm. One of the important steps in this algorithm is to determine the optimal number of clusters into which the data may be clustered. From the given graph find the value of K that represents the number of clusters:



☐ 1

☐ 2

☐ 3

Question 15

Max. score: 4.00

In Machine Learning, you are working on data processing and want to handle outliers by using the univariate method. In this scenario, which of the following statements about the univariate method are correct:

1. It looks for unusual combinations in all the variables.
2. It looks for data points having extreme values on a single variable.
3. It uses box plots for detecting outliers.
4. It reduces the contribution of potential outliers in the training process.

☐ 1 and 2

☐ 2 and 3

☐ 3 and 4

☐ 1 and 4

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Question 16

Max. score: 4.00

In Machine Learning, which of the following statements about networks is correct?

☐ It is possible to train a network by initializing all the weights to zero.

☐ It is possible to train a network by initializing all the biases to zero.

☐ Both of these

☐ None of these

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Question 17

Max. score: 4.00

In Machine Learning, the R-squared method is used to measure the goodness-of-fit feature of models. A feature is added to the linear regression model to train the same model again. Which of the following statements about the R-squared value is correct?

☐ If the R-Squared value increases, then the variable of the model remains significant.

☐ If the R-Squared value decreases, then the variable of the model remains insignificant.

☐ The R-Squared value cannot determine the variance of a model.

☐ None of these

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Question 18

Max. score: 4.00

John is working on a binary classification problem and needs help understanding what it means if his AUC score is nearly zero while working on an application. Can you help him find the appropriate information?

☐ Model is reciprocating the result.

☐ Model is performing well.

☐ Model is overfitting.

☐ None of these

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Question 19

Max. score: 4.00

In Machine Learning, you are working on a project using natural language processing. You want to implement the continuous bag of words model. In this scenario, which of the following statements about the continuous bag of words model are correct:

Statement:

1. It is used to predict the word in the middle.
2. It is used to predict the context.
3. It finds the probability of a word occurring in a context.
4. The model architecture uses neural networks.

☐ 1, 2, and 3

☐ 2, 3, and 4

☐ 1, 3, and 4

☐ All of these

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Question 20

Max. score: 4.00

John is given the eight actual values of the **target** variable in the **train** file of a Machine Learning model. The values are as follows:

[0, 0, 0, 0, 1, 0, 1, 1]

According to the given values, what is the entropy of the target variable?

☐ $-(5/8 \log(5/8) + 3/8 \log(3/8))$

☐ $(5/8 \log(5/8) - 3/8 \log(3/8))$

☐ $(5/8 \log(3/8) - 3/8 \log(5/8))$

☐ $-(5/8 \log(3/8) + 3/8 \log(5/8))$

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Question 21

Max. score: 20.00

0.3)

Auto-complete ready! Autosaved

Servers

A company has N servers, labeled from 1 to N . In the beginning, all the servers have no workload.

M requests are sent to the company's API. The company will distribute these requests evenly among its servers to keep the workload balanced. The workload is balanced when the busiest server has the least possible load.

Which server will handle the request for each API request made?

Notes

- Each request adds to the overall workload of the servers.
- If two servers have the same minimum workload, the server that is labeled with the smaller number will handle the request.

Function description

Complete the `solve()` function provided in the editor. The function takes the following 3 parameters and returns the solution:

- N : Represents the number of servers
- M : Represents the number of requests
- `requests`: Represents the load of each request

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't

```
1 > #include<stdio.h>--
4 int* solution (int N, int M, int* requests, int*
  out_n) {
5     // Write your code here
6     // Assign the size of the array to be returned
  into out_n variable
7     // *out_n = array_size;
8     int* arr = (int*)malloc(N * sizeof(int));
9     return arr;
10 > }--
```

request.

Function description

Complete the *solve()* function provided in the editor. The function takes the following 3 parameters and returns the solution:

- *N*: Represents the number of servers
- *M*: Represents the number of requests
- *requests*: Represents the load of each request

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains *N* denoting the number of servers.
- The second line contains *M* denoting the number of requests.
- The third line contains an array *requests* denoting the load of each request.

Output format

Print an array, denoting the server that serves each request.

Constraints

$$1 \leq N \leq 10^5$$

$$1 \leq M \leq 10^5$$

$$1 \leq \text{request}[i] \leq 10^9$$

```
1 > #include<stdio.h>--
4 int* solution (int N, int M, int* requests, int*
  out_n) {
5     // Write your code here
6     // Assign the size of the array to be returned
  into out_n variable
7     // *out_n = array_size;
8     int* arr = (int*)malloc(N * sizeof(int));
9     return arr;
10 > }-
```

Sample input

```
2
2
5 5
```

Sample output

```
1 2
```

Explanation

Given

Input

- $N = 2$
- $M = 2$
- $requests = [5, 5]$

Output: 1 2

Approach

- When the first request is made, both servers have the same load so server 1 serves.
- When the second request is made, if server 2 serves then the maximum load is 5 else it is 10. Hence, server 2 serves.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden

```
1 > #include<stdio.h>--
4 int* solution (int N, int M, int* requests, int*
  out_n) {
5     // Write your code here
6     // Assign the size of the array to be returned
  into out_n variable
7     // *out_n = array_size;
8     int* arr = (int*)malloc(N * sizeof(int));
9     return arr;
10 > }--
```

Question 22

Max. score: 20.00

Consistent data

A sequence containing only 0s and 1s is called a consistent sequence if no adjacent values in the sequence are the same. For example (010101....) is a consistent sequence while (011) is not. You are allotted a simple task.

Given an integer N that represents the size of a consistent sequence starting with '0'.

Find out the number of consistent subsequences of the given sequence mod $10^9 + 7$.

Note: A subsequence is a sequence that can be derived from the given sequence by deleting zero or more elements without changing the order of the remaining elements.

Function description

Complete the function `Consistent()`. This function takes the following parameter and returns the required answer:

- N : Represents the size of the sequence

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains an integer N representing the sequence's size.

Output format

Print the number of consistent subsequences mod $10^9 + 7$.

```
1 > #include<stdio.h>--
4 int Consistent (int N) {
5     // Write your code here
6     return 0;
7 > }--
```

$1 \leq N \leq 10^6$

Sample input

3

Sample output

6

Explanation

Assumption

- $N = 3$

Approach

- The given consistent sequence is [0, 1, 0]
- The number of consistent subsequences is 6 which are {[0], [1], [0], [0, 1], [1, 0], [0, 1, 0]} for which the corresponding indices sets are {[1], [2], [3], [1, 2], [2, 3], [1, 2, 3]}.

Therefore we will return 6 as our answer.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

```
1 >#include<stdio.h>--
4 int Consistent (int N) {
5     // Write your code here
6     return 0;
7 >}--
```


Question 23

Max. score: 20.00

One block

You are working in the resource distribution team of your company. A *One block* is a block of data having exactly one resource which has value 1. You are given an array *Arr* containing *N* resource values.

What is the number of ways to divide the array into continuous blocks such that each block is *One block*.

NOTE - If we cannot form a block with *One block*, then output 0.

Function description

Complete the function `OneBlock()`. This function takes the following 2 parameters and returns the required answer:

- *N*: Represents the number of resources
- *Arr*: Represents the value of resources

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains an integer *N* denoting the number of resources.
- The second line contains *N* space-separated integers denoting the value of resources.

Output format

Print the number of ways to divide the array into continuous blocks such that each block is *One block*.

```
1 > #include<stdio.h>--
4 long long OneBlock (int N, int* Arr) {
5     // Write your code here
6
7     return 0;
8 > }--
```

Output format

Print the number of ways to divide the array into continuous blocks such that each block is *One block*.

Constraints

$$1 \leq N \leq 100$$

$$0 \leq Arr_i \leq 1$$

Sample input

```
3
0 1 0
```

Sample output

```
1
```

Explanation

Given

- $N = 3$
- $Arr = [0, 1, 0]$

Approach

There is exactly one resource that has the value of 1. Hence, the number of ways is 1.

```
1 >#include<stdio.h>--
4 long long OneBlock (int N, int* Arr) {
5     // Write your code here
6
7     return 0;
8 >}-
```

Explanation

Given

- $N = 3$
- $Arr = [0, 1, 0]$

Approach

There is exactly one resource that has the value of 1. Hence, the number of ways is 1.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 1.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Perl, C, C++, C#, Clojure, C++, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, Java 17, JavaScript(Node.js), Julia, Kotlin, Lisp

```
1 > #include<stdio.h>--
4 long long OneBlock (int N, int* Arr) {
5     // Write your code here
6
7     return 0;
8 > }--
```

Question 22

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Consistent data

A sequence containing only 0s and 1s is called a consistent sequence if no adjacent values in the sequence are the same. For example (010101....) is a consistent sequence while (011) is not. You are allotted a simple task.

Given an integer N that represents the size of a consistent sequence starting with '0'.

Find out the number of consistent subsequences of the given sequence mod $10^9 + 7$.

Note: A subsequence is a sequence that can be derived from the given sequence by deleting zero or more elements without changing the order of the remaining elements.

Function description

Complete the function `Consistent()`. This function takes the following parameter and returns the required answer:

- N : Represents the size of the sequence

Input format for custom testing

Note: Use this input format if you are testing against custom input or writing code in a language where we don't provide boilerplate code

- The first line contains an integer N representing the sequence's size.

Output format

Print the number of consistent subsequences mod $10^9 + 7$.

C++17 (g++ 10.3.0)

Auto-complete ready! Autosaved

```
1 > #include<bits/stdc++.h>-
3 int Consistent (int N) {
4     int MOD = 1e9 + 7;
5
6     return (((N%MOD) * (N+1)%MOD) / 2)%MOD;
7 > }-
```

Visible test cases Custom Input **Result**

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Test case 4 **Wrong answer**

Input

5

Output

15

Expected output

19

Question 1

Max. score: 2.00

Anita built a machine learning model that initially worked smoothly. However, the model suddenly suffered from high variance and faced low bias. Which supervised learning algorithm can she use in this scenario to tackle high variance?

☐ Classification

☐ Linear Regression

☐ Naïve Bayes

☐ Random Forest

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John is working on Machine Learning. While working on an application he wants to know the cost function for using logistic regression modeling. Help him to find which of the following can be used in the above-mentioned context.

Options:

1. $\frac{-1}{m} [\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (y^{(i)}) \log(1 - h\theta(x^{(i)}))]$
2. $\frac{-1}{m} [\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h\theta(x^{(i)}))]$
3. $\frac{-1}{m} [\sum_{i=1}^m y^{(i)} \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(h\theta(x^{(i)}))]$
4. $\frac{-1}{m} [\sum_{i=1}^m \log(h\theta(x^{(i)})) + (1 - y^{(i)}) \log(1 - h\theta(x^{(i)}))]$

☐ 1

☐ 2

☐ 3

☒ 4

Question 3

Max. score: 4.00

A company wants to track the number of calls they receive over a day to decide how many people they want to hire. They make the following assumptions.

You are given a task to determine the probability of receiving exactly k number of calls during a specific day. Which of the following distributions can you use to complete this task:

Assumptions

1. Each call occurs independently from every other call.
2. A given call has the same probability of occurring at any given time of the day.
3. Calls occur at a rate of λ calls per day.

☐ Binomial

☐ Poisson

☐ Normal

☐ None of these

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Question 4

Max. score: 4.00

Which of the following statements about the **convolution** layer are correct:

Statements:

1. This is the layer where a majority of the computations occur. Here, the computer distinguishes between the white pixels as 1 and black pixels as -1.
2. This is the layer where a majority of the computations occur. Here, the computer distinguishes between the white pixels as 0 and black pixels as 1.
3. When you input an image through this layer, a model extracts the unique feature of an image such as colors, shapes, and edges that defines the image. Once you have the features of the image, you perform the filtering process.
4. This layer produces multiple feature maps over different iterations.

☐ 1, 3, and 4

☐ 2, 3, and 4

☐ 1 and 3

☐ 2 and 4

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Question 5

Max. score: 4.00

In Machine Learning, which of the following statements about PCA and LDA is correct:

1. PCA is non-linear and LDA is linear
2. PCA is supervised and LDA is unsupervised

☐ 1

☐ 2

☐ Both of these

☒ None of these

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Question 6

Max. score: 4.00

In Machine Learning, which of the following statements about L1 and L2 regularizations is correct?

Statements:

1. The L2 regularization tends to spread error among all the terms, while L1 regularization is binary in nature, with many variables either being assigned 1 or 0 in weighting.
2. The L1 and L2 regularization correspond to set a Laplacian and Gaussian prior respectively to the terms.

☐ 1

☐ 2

☐ Both of these

☐ None of these

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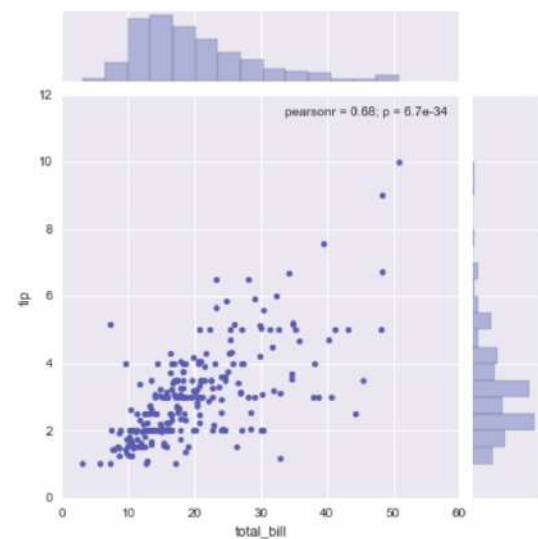
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Question 7

Max. score: 4.00

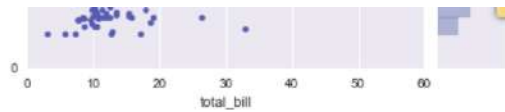
Tara is given the following graph. Which of the following statements about this graph are correct?

Graph:



Statements:

1. This graph represents a scatter plot and denotes a linear relationship between tip and total_bill.
2. This graph represents a joint plot and denotes a linear relationship between tip and total_bill.



Statements:

1. This graph represents a scatter plot and denotes a linear relationship between tip and total_bill.
2. This graph represents a joint plot and denotes a linear relationship between tip and total_bill.
3. The tip is left-skewed, whereas total_bill is right-skewed.
4. The tip is right-skewed, whereas total_bill is left-skewed.

☐ 1 and 2

☐ 1 and 3

☐ 2 and 3

☐ 2 and 4

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Question 8

Max. score: 6.00

John has the confusion matrix for the classification model. It contains information about the actual and predicted classification of patients either having the disease or not having the disease (Yes/No). Calculate sensitivity, precision, negative predicted value, and accuracy for his model.

Image

		Predicted Class	
		Yes	No
Actual Class	Yes	40	25
	No	17	18

☐ Sensitivity = 0.62, precision = 0.70, negative predicted value = 0.41, and accuracy = 0.58

☐ Sensitivity = 0.70, precision = 0.58, negative predicted value = 0.62, and accuracy = 0.41

☐ Sensitivity = 0.58, precision = 0.41, negative predicted value = 0.70, and accuracy = 0.62

☐ Sensitivity = 0.41, precision = 0.62, negative predicted value = 0.58, and accuracy = 0.70

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Question 9

Max. score: 2.00

In Machine Learning, you are working on a project using natural language processing (NLP). You are using n-gram in the sentence, "I want to code". Which of the following options is the bigram of the provided sentence?

Options

1. I want want to to code
2. I want to code
3. I want to want to code
4. I want to code

☐ 1

☐ 2

☐ 3

☐ 4

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